Kafka Topics Architecture

Kafka versus MOM

Kafka vs JMS, SQS, RabbitMQ Messaging

Tos

- Is Kafka a Queue or a Pub/Sub/Topic?
  - Yes
- Kafka is like a Queue per consumer group
  - Kafka is a queue system per consumer in consumer group so load balancing like JMS, RabbitMQ queue
- Kafka is like Topics in JMS, RabbitMQ, MOM
  - Topic/pub/sub by offering Consumer Groups which act like subscriptions
  - Broadcast to multiple consumer groups
- MOM = JMS, ActiveMQ, RabbitMQ, IBM MQ Series, Tibco, etc.

Kafka vs MOM

Tos

- By design, Kafka is better suited for scale than traditional MOM systems due to partition topic log
  - Load divided among Consumers for read by partition
  - Handles parallel consumers better than traditional MOM
- Also by moving location (partition offset) in log to client/consumer side of equation instead of the broker, less tracking required by Broker and more flexible consumers
- Kafka written with mechanical sympathy, modern hardware, cloud in mind
  - Disks are faster
  - Servers have tons of system memory
  - Easier to spin up servers for scale out

Topics, Logs, Partitions

Tos

- Kafka ***Topic*** is a stream of records
- ***Topics*** stored in log
- ***Log*** broken up into ***partitions*** and ***segments***
- ***Topic*** is a category or stream name or feed
- Topics are pub/sub
  - Can have zero or many subscribers - consumer groups
- ***Topics*** are broken up and spread by partitions for speed and size

Topic Partitions

Tos

- ***Topics*** are broken up into ***partitions***
- ***Partitions*** decided usually by key of record
  - Key of record determines which partition
- ***Partitions*** are used to scale Kafka across many servers
  - Record sent to correct partition by key
- ***Partitions*** are used to facilitate parallel consumers
  - Records are consumed in parallel up to the number of partitions
- Order guaranteed per partition
- Partitions can be ***replicated*** to multiple brokers

Topic Partition Log

Tos

- ***Order*** is maintained only in a single ***partition***
  - ***Partition*** is ordered, immutable sequence of records that is continually appended to—a structured commit ***log***
- ***Records*** in partitions are assigned ***sequential id*** number called the *offset*
- ***Offset*** identifies each record within the partition
- ***Topic Partitions*** allow Kafka log to scale beyond a size that will fit on a single server
  - Topic partition must fit on servers that host it
  - topic can span many partitions hosted on many servers

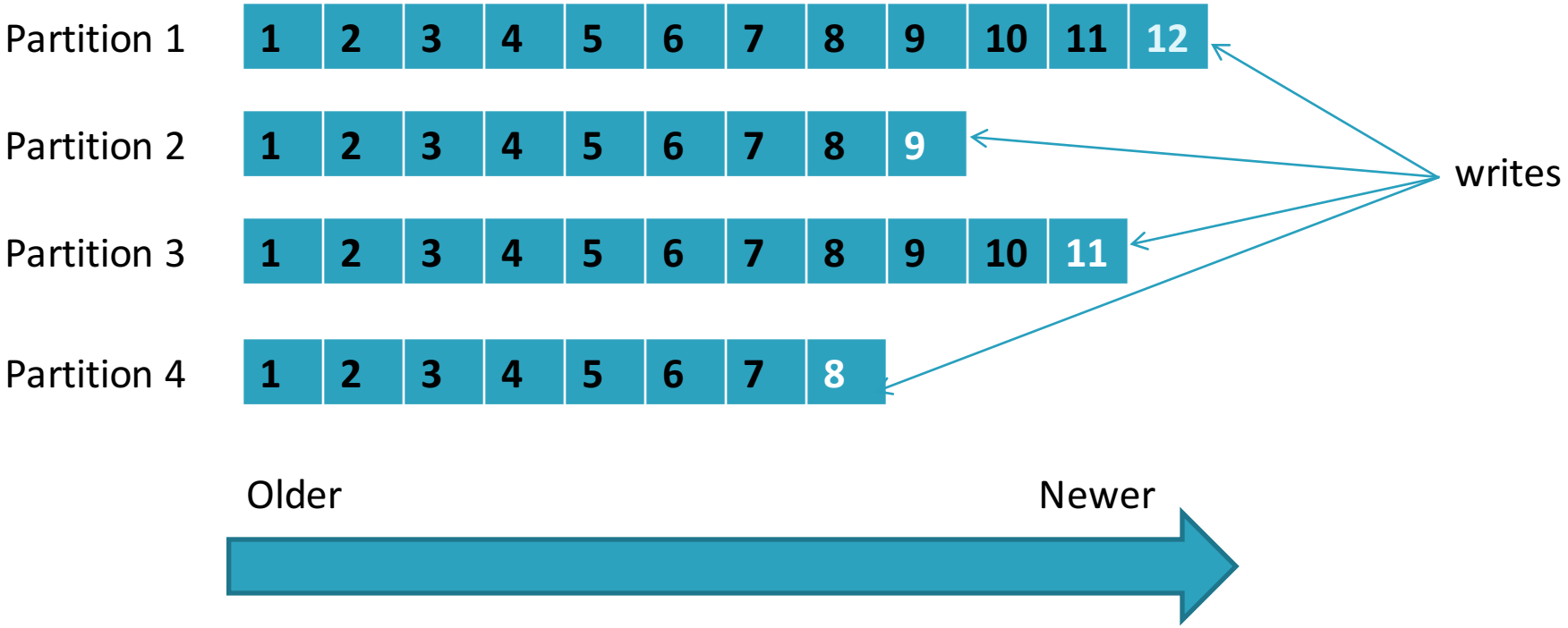
Topic Parallelism and consumers

Tos

- Topic Partitions are unit of ***parallelism***
  partition can only be used by one consumer in group at a time
- Consumers can run in their own process or their own thread
- If a consumer stops, Kafka spreads partitions across remaining consumer in group
- #of Consumers you can run per Consumer Group limited by #of Partitions
- Consumers getting assigned partition aids in efficient message consumption tracking

Kafka Topic Partitions Layout

Tos



Kafka Scale and Speed

Tos

- How can Kafka scale if multiple producers and consumers read/write to same Kafka Topic log?
- Writes fast: Sequential writes to file system are ***fast*** (700 MB or more a second)
- Scales writes and reads by ***sharding:***
  - Topic logs into ***Partitions*** (parts of a Topic log)
  - Topics logs can be split into multiple Partitions ***different machines/different disks***
  - Multiple Producers can write to different Partitions of the same Topic
  - Multiple Consumers Groups can read from different partitions

Replication: Kafka Partition Distribution

Tos

- Each partition has ***leader server*** and zero or more ***follower servers***
  - ***Leader*** handles all read and write requests for partition
  - ***Followers*** replicate leader, and take over if leader dies
  - Used for parallel consumer handling within a group
- Partitions of log are distributed over the servers in the Kafka cluster with each server handling data and requests for a share of partitions
- Each partition can be replicated across a configurable number of Kafka servers - Used for fault tolerance

Replication: Kafka Partition leader

Tos

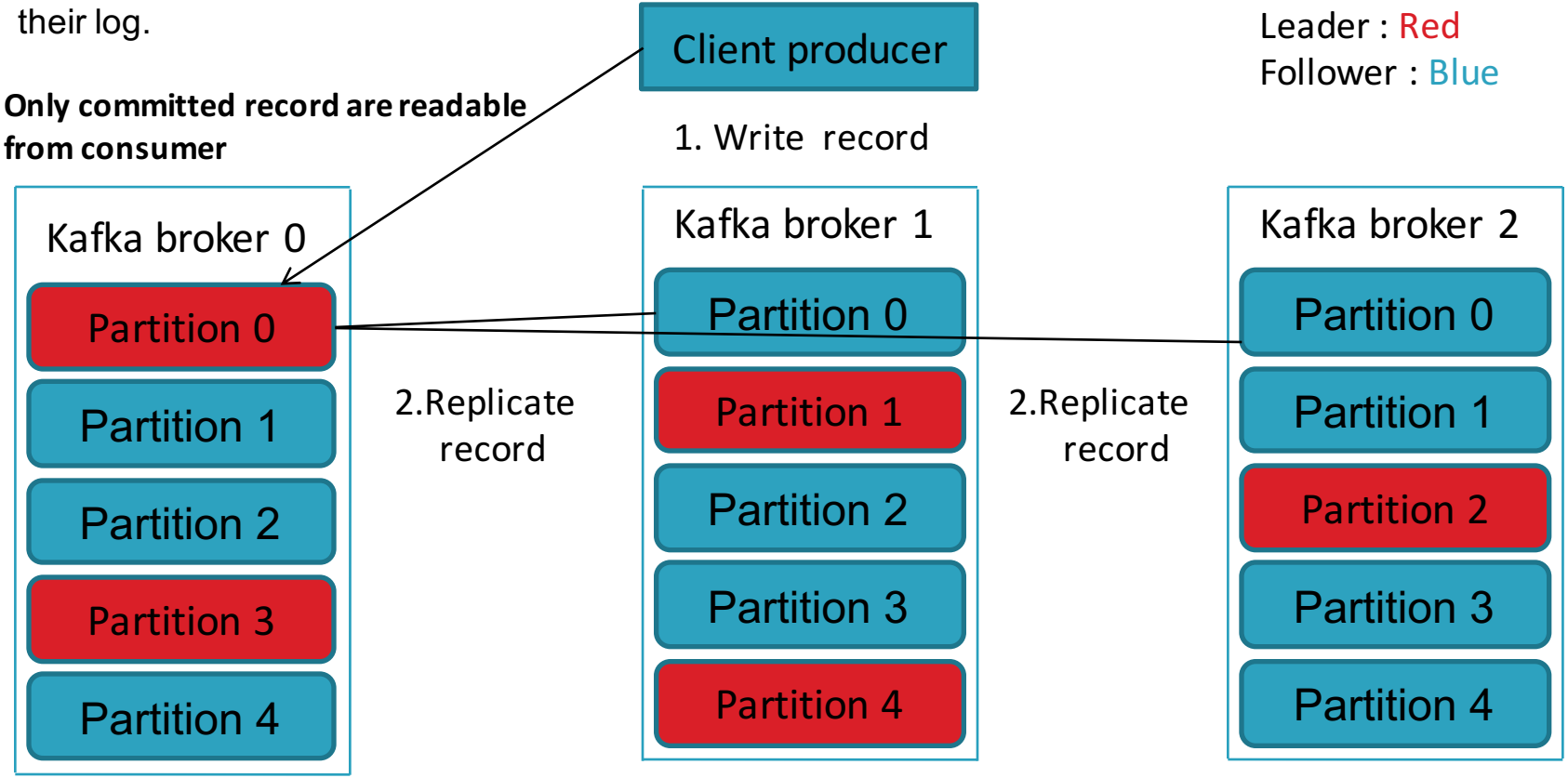
- One node/partition's replicas is chosen as ***leader***
- Leader handles all reads and writes of Records for partition
- Writes to partition are ***replicated*** to ***followers*** (node/partition pair)
- An ***follower*** that is ***in-sync*** is called an ***ISR (in-sync replica)***
- If a partition leader fails, one ISR is chosen as new leader

Kafka Replication to Partition 0

Tos

Record is considered "committed". When all ISRs for partition wrote to their log.

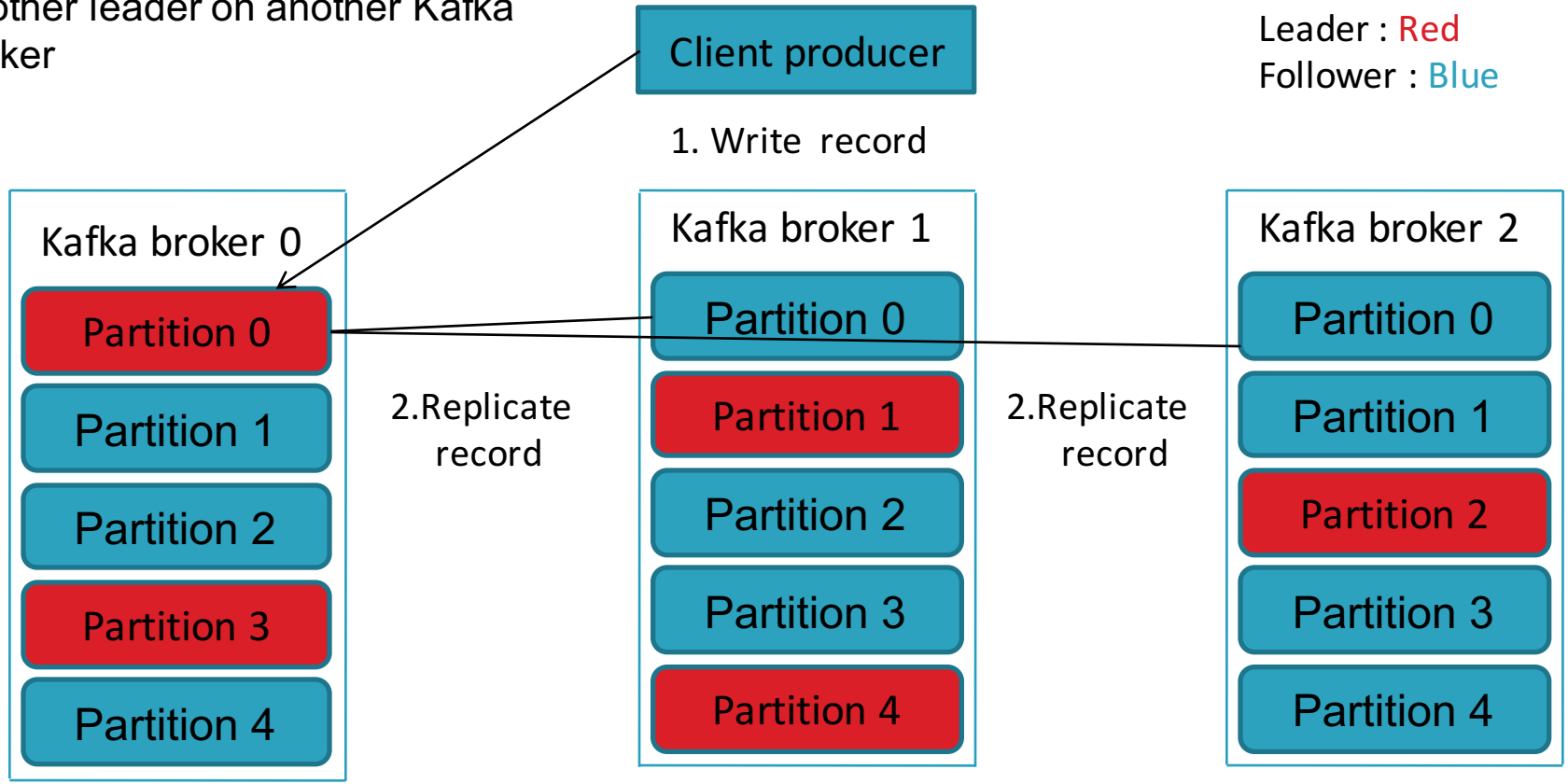
Only committed record are readable from consumer



Kafka Replication to Partition 1

Tos

Another partition can be owned by another leader on another Kafka broker



Run Kafka

Tos

- Run Zoo Keeper start up script
- Run Kafka Server/Broker start up script
- Create Kafka Topic from command line

Run ZooKeeper

Tos

run-zookeeper.sh

```
#!/usr/bin/env bash
cd ~/kafka-training

kafka/bin/zookeeper-server-start.sh \
   kafka/config/zookeeper.properties

```

```
$ ./run-zookeeper.sh
[2017-05-13 13:34:52,489] INFO Reading configuration from: kafka/config/zookeeper.properties (org.
apache.zookeeper.server.quorum.QuorumPeerConfig)
[2017-05-13 13:34:52,491] INFO autopurge.snapRetainCount set to 3 (org.apache.zookeeper.server.Dat
adirCleanupManager)
[2017-05-13 13:34:52,491] INFO autopurge.purgeInterval set to 0 (org.apache.zookeeper.server.Datad
irCleanupManager)
[2017-05-13 13:34:52,491] INFO Purge task is not scheduled. (org.apache.zookeeper.server.DatadirCl
eanupManager)
[2017-05-13 13:34:52,491] WARN Either no config or no quorum defined in config, running  in standa
lone mode (org.apache.zookeeper.server.quorum.QuorumPeerMain)
[2017-05-13 13:34:52,504] INFO Reading configuration from: kafka/config/zookeeper.properties (org.
apache.zookeeper.server.quorum.QuorumPeerConfig)
[2017-05-13 13:34:52,504] INFO Starting server (org.apache.zookeeper.server.ZooKeeperServerMain)
[2017-05-13 13:34:57,609] INFO Server environment:zookeeper.version=3.4.9-1757313, built on 08/23/
2016 06:50 GMT (org.apache.zookeeper.server.ZooKeeperServer)
[2017-05-13 13:34:57,609] INFO Server environment:host.name=10.0.0.115 (org.apache.zookeeper.serve
```

Run Kafka Server

Tos

run-kafka.sh

```
#!/usr/bin/env bash
cd ~/kafka-training

kafka/bin/kafka-server-start.sh \
    kafka/config/server.properties
```

```
$ ./run-kafka.sh
[2017-05-13 13:47:01,497] INFO KafkaConfig values:
        advertised.host.name = null
        advertised.listeners = null
        advertised.port = null
        authorizer.class.name =
        auto.create.topics.enable = true
        auto.leader.rebalance.enable = true
        background.threads = 10
        broker.id = 0
        broker.id.generation.enable = true
        broker.rack = null
        compression.type = producer
        connections.max.idle.ms = 600000
        controlled.shutdown.enable = true
        controlled.shutdown.max.retries = 3
        controlled.shutdown.retry.backoff.ms = 5000
        controller.socket.timeout.ms = 30000
```

Create Kafka Topic

Tos

create-topic.sh

```bash
#!/usr/bin/env bash

cd ~/kafka-training

# Create a topic
kafka/bin/kafka-topics.sh --create --zookeeper localhost:2181 \
--replication-factor 1 --partitions 13 --topic my-topic
```

```
$ ./create-topic.sh
Created topic "my-topic".
```

List Topics

Tos

list-topics.sh

```
#!/usr/bin/env bash

cd ~/kafka-training

# List existing topics
kafka/bin/kafka-topics.sh --list \
    --zookeeper localhost:2181

```

```
~/kafka-training/lab1/solution
$ ./list-topics.sh
__consumer_offsets
_schemas
my-example-topic
my-example-topic2
my-topic
new-employees
```

Lab - Basic Kafka Operations - CLI (Topic) – 30 Mins